

CHAPTER 16 Generation–Recombination

This chapter describes the generation–recombination processes that can be modeled in Sentaurus Device.

Generation–recombination processes are processes that exchange carriers between the conduction band and the valence band. They are very important in device physics, in particular, for bipolar devices. This chapter describes the generation–recombination models available in Sentaurus Device. Most models are local in the sense that their implementation (sometimes in contrast to reality) does not involve spatial transport of charge. For each individual generation or recombination process, the electrons and holes involved appear or vanish at the same location. The only exceptions are one of the trap-assisted tunneling models and one of the band-to-band tunneling models (see [Dynamic Nonlocal Path Trap-assisted Tunneling on page 423](#) and [Dynamic Nonlocal Path Band-to-Band Model on page 454](#)). For other models that couple the conduction and valence bands and account for spatial transport of charge, see [Tunneling and Traps on page 478](#) and [Nonlocal Tunneling at Interfaces, Contacts, and Junctions on page 710](#).

Shockley–Read–Hall Recombination

Recombination through deep defect levels in the gap is usually labeled Shockley–Read–Hall (SRH) recombination. In Sentaurus Device, the following form is implemented:

$$R_{\text{net}}^{\text{SRH}} = \frac{np - n_{\text{i,eff}}^2}{\tau_p(n + n_1) + \tau_n(p + p_1)} \quad (355)$$

with:

$$n_1 = n_{\text{i,eff}} \exp\left(\frac{E_{\text{trap}}}{kT}\right) \quad (356)$$

and:

$$p_1 = n_{\text{i,eff}} \exp\left(\frac{-E_{\text{trap}}}{kT}\right) \quad (357)$$

where E_{trap} is the difference between the defect level and intrinsic level. The variable E_{trap} is accessible in the parameter file. The silicon default value is $E_{\text{trap}} = 0$.

The lifetimes τ_n and τ_p are modeled as a product of a doping-dependent (see [SRH Doping Dependence on page 417](#)), field-dependent (see [SRH Field Enhancement on page 419](#)), and temperature-dependent (see [SRH Temperature Dependence on page 418](#)) factor:

$$\tau_c = \tau_{\text{dop}} \frac{f(T)}{1 + g_c(F)} \quad (358)$$

where $c = n$ or $c = p$. For an additional density dependency of the lifetimes, see [Trap-assisted Auger Recombination on page 427](#).

For simulations that use Fermi statistics (see [Fermi Statistics on page 220](#)) or quantization (see [Chapter 14 on page 315](#)), [Eq. 355](#) needs to be generalized. The modified equation reads:

$$R_{\text{net}}^{\text{SRH}} = \frac{np - \gamma_n \gamma_p n_i^2}{\tau_p (n + \gamma_n n_1) + \tau_n (p + \gamma_p p_1)} \quad (359)$$

where γ_n and γ_p are given by [Eq. 49](#) and [Eq. 50, p. 221](#).

Using SRH Recombination

The generation–recombination models are selected in the `Physics` section as an argument to the `Recombination` keyword:

```
Physics{ Recombination( <arguments> ) ... }
```

The SRH model is activated by specifying the `SRH` argument:

```
Physics{ Recombination( SRH ... ) ... }
```

The keyword for plotting the SRH recombination rate is:

```
Plot{ ...
    SRHRecombination
}
```

SRH Doping Dependence

The doping dependence of the SRH lifetimes is modeled in Sentaurus Device with the Scharfetter relation:

$$\tau_{\text{dop}}(N_{A,0} + N_{D,0}) = \tau_{\text{min}} + \frac{\tau_{\text{max}} - \tau_{\text{min}}}{1 + \left(\frac{N_{A,0} + N_{D,0}}{N_{\text{ref}}} \right)^{\gamma}} \quad (360)$$

Such a dependence arises from experimental data [1] and the theoretical conclusion that the solubility of a fundamental, acceptor-type defect (probably a divacancy (E5) or a vacancy complex) is strongly correlated to the doping density [2][3][4]. Default values are listed in [Table 71 on page 419](#). The Scharfetter relation is used when the argument `DopingDependence` is specified for the SRH recombination. Otherwise, $\tau = \tau_{\text{max}}$ is used.

The evaluation of the SRH lifetimes according to the Scharfetter model is activated by specifying the additional argument `DopingDependence` for the SRH keyword in the Recombination statement:

```
Physics{ Recombination( SRH( DopingDependence ... ) ... ) ... }
```

Lifetime Profiles From Files

Sentaurus Device can use spatial lifetime profiles provided by a file. Such profiles can be precomputed or generated manually by an editor such as Sentaurus Structure Editor (refer to the *Sentaurus™ Structure Editor User Guide*).

The names of the datasets for the electron and hole lifetimes must be `eLifetime` and `hLifetime`, respectively. They are loaded from a file named by the keyword `LifeTime` in the `File` section:

```
File {
    Grid      = "MyDev_msh.tdr"
    LifeTime  = "MyDev_msh.tdr"
    ...
}
```

For each grid point, the values defined by the lifetime profile are used as τ_{max} in [Eq. 360](#).

The lifetime data can be in a separate file from the doping, but must correspond to the same grid.

SRH Temperature Dependence

To date, there is no consensus on the temperature dependence of the SRH lifetimes. This appears to originate from a different understanding of lifetime. From measurements of the recombination lifetime [5][6]:

$$\tau = \delta n / R \quad (361)$$

in power devices (δn is the excess carrier density under neutral conditions, $\delta n = \delta p$), it was concluded that the lifetime increases with rising temperature. Such a dependence was modeled either by a power law [5][6]:

$$\tau(T) = \tau_0 \left(\frac{T}{300 \text{ K}} \right)^\alpha \quad (362)$$

or an exponential expression of the form:

$$\tau(T) = \tau_0 e^{c \left(\frac{T}{300 \text{ K}} - 1 \right)} \quad (363)$$

A calculation using the low-temperature approximation of multiphonon theory [7] gives:

$$\tau_{\text{SRH}}(T) = \tau_{\text{SRH}}(300 \text{ K}) \cdot f(T) \quad \text{with} \quad f(T) = \left(\frac{T}{300 \text{ K}} \right)^{T_\alpha} \quad (364)$$

with $T_\alpha = -3/2$, which is the expected decrease of minority carrier lifetimes with rising temperature. Since the temperature behavior strongly depends on the nature of the recombination centers, there is no universal law $\tau_{\text{SRH}}(T)$.

In Sentaurus Device, the power law model, Eq. 362, can be activated with the keyword TempDependence in the SRH statement:

```
Physics{ Recombination( SRH( TempDependence ...) ... ) }
```

Additionally, Sentaurus Device supports an exponential model for $f(T)$:

$$f(T) = e^{c \left(\frac{T}{300 \text{ K}} - 1 \right)} \quad (365)$$

This model is activated with the keyword ExpTempDependence:

```
Physics{ Recombination( SRH( ExpTempDependence ...) ... ) }
```

SRH Doping- and Temperature-dependent Parameters

All the parameters of the doping- and temperature-dependent SRH recombination models are accessible in the parameter set `Scharfetter`.

Table 71 Default parameters for doping- and temperature-dependent SRH lifetime

Symbol	Parameter name	Electrons	Holes	Unit
$\tau_{\min}$	taumin	0	0	s
$\tau_{\max}$	taumax	1×10^{-5}	3×10^{-6}	s
N_{ref}	Nref	1×10^{16}	1×10^{16}	cm^{-3}
γ	gamma	1	1	1
T_{α}	Talpha	–1.5	–1.5	1
C	Tcoeff	2.55	2.55	1
E_{trap}	Etrap	0	0	eV

SRH Field Enhancement

Field enhancement reduces SRH recombination lifetimes in regions of strong electric fields. It must not be neglected if the electric field exceeds a value of approximately $3 \times 10^5 \text{ V/cm}$ in certain regions of the device. For example, the I–V characteristics of reverse-biased p–n junctions are extremely sensitive to defect-assisted tunneling, which causes electron–hole pair generation before band-to-band tunneling or avalanche generation sets in. Therefore, it is recommended that field-enhancement is included in the simulation of drain reverse leakage and substrate currents in MOS transistors.

Sentaurus Device provides two field-enhancement models: the Schenk trap-assisted tunneling model (see [Schenk Trap-assisted Tunneling \(TAT\) Model on page 420](#)) and the Hurkx trap-assisted tunneling model (see [Hurkx TAT Model on page 422](#)). The Hurkx model is also available for trap capture and emission rates (see [Hurkx Model for Cross Sections on page 474](#)).

Using Field Enhancement

The local field-dependence of the SRH lifetimes is activated by parameters in the `ElectricField` option of SRH:

```
SRH( ...  
    ElectricField (  
        Lifetime = Schenk | Hurkx | Constant  
        DensityCorrection = Local | None  
    )  
)
```

`Lifetime` selects the lifetime model. The default is `Constant`, for field-independent lifetime. `Schenk` selects the Schenk lifetimes (see [Schenk Trap-assisted Tunneling \(TAT\) Model on page 420](#)), and `Hurkx` selects the Hurkx lifetimes (see [Hurkx TAT Model on page 422](#)).

`DensityCorrection` defaults to `None`. A value of `Local` selects the model described in [Schenk TAT Density Correction on page 422](#).

For backward compatibility:

- `SRH (Tunneling)` selects `Lifetime = Schenk` and `DensityCorrection = Local`.
- `SRH(Tunneling(Hurkx))` selects `Lifetime = Hurkx` and `DensityCorrection = None`.

NOTE The inclusion of defect-assisted tunneling may lead to convergence problems. In such cases, it is recommended that the flag `NoSRHperPotential` is specified in the `Math` section. This causes Sentaurus Device to exclude derivatives of $g(F)$ with respect to the potential from the Jacobian matrix.

Schenk Trap-assisted Tunneling (TAT) Model

The field dependence of the recombination rate is taken into account by the field enhancement factors:

$$[1 + g(F)]^{-1} \tag{366}$$

of the SRH lifetimes [\[7\]](#) (see [Eq. 358](#)).

In the case of electrons, $g(F)$ has the form:

$$g_n(F) = \left(1 + \frac{(\hbar\Theta)^{3/2} \sqrt{E_t - E_0}}{E_0 \hbar\omega_0} \right)^{-\frac{1}{2}} \frac{(\hbar\Theta)^{3/4} (E_t - E_0)^{1/4}}{2 \sqrt{E_t E_0}} \left(\frac{\hbar\Theta}{kT} \right)^{\frac{3}{2}} \quad (367)$$

$$\times \exp \left(-\frac{E_t - E_0}{\hbar\omega_0} + \frac{\hbar\omega_0 - kT}{2\hbar\omega_0} + \frac{2E_t + kT}{2\hbar\omega_0} \ln \frac{E_t}{\epsilon_R} - \frac{E_0}{\hbar\omega_0} \ln \frac{E_0}{\epsilon_R} + \frac{E_t - E_0}{kT} - \frac{4}{3} \left(\frac{E_t - E_0}{\hbar\Theta} \right)^{\frac{3}{2}} \right)$$

where E_0 denotes the energy of an optimum horizontal transition path, which depends on field strength and temperature in the following way:

$$E_0 = 2 \sqrt{\epsilon_F [\sqrt{\epsilon_F + E_t + \epsilon_R} - \sqrt{\epsilon_F}] - \epsilon_R}, \epsilon_F = \frac{(2\epsilon_R kT)^2}{3(\hbar\Theta)} \quad (368)$$

In this expression, $\epsilon_R = S\hbar\omega_0$ is the lattice relaxation energy, S is the Huang–Rhys factor, $\hbar\omega_0$ is the effective phonon energy, E_t is the energy level of the recombination center (thermal depth), and $\Theta = (q^2 F^2 / 2\hbar m_{\Theta,n})^{1/3}$ is the electro-optical frequency. The mass $m_{\Theta,n}$ is the electron tunneling mass in the field direction and is given in the parameter file. The expression for holes follows from Eq. 367 by replacing $m_{\Theta,n}$ with $m_{\Theta,p}$ and E_t with $E_{g,\text{eff}} - E_t$.

For electrons, E_t is related to the defect level E_{trap} of Eq. 356 and Eq. 357 by:

$$E_t = \frac{1}{2} E_{g,\text{eff}} + \frac{3}{4} kT \ln \left(\frac{m_n}{m_p} \right) - E_{\text{trap}} - (32 R_C \hbar^3 \Theta^3)^{1/4} \quad (369)$$

where the effective Rydberg constant R_C is:

$$R_C = m_C \left(\frac{Z^2}{\epsilon^2} \right) Ry \quad (370)$$

where Ry is the Rydberg energy (13.606 eV), ϵ is the relative dielectric constant, and Z is a fit parameter.

For holes, E_t is given by:

$$E_t = \frac{1}{2} E_{g,\text{eff}} - \frac{3}{4} kT \ln \left(\frac{m_n}{m_p} \right) + E_{\text{trap}} - (32 R_V \hbar^3 \Theta^3)^{1/4} \quad (371)$$

Note that E_{trap} is measured from the intrinsic level and not from mid gap. The zero-field lifetime τ_{SRH} is defined by Eq. 360.

Schenk TAT Density Correction

In the original Schenk model [7], the density n in Eq. 355 is replaced by:

$$\tilde{n} = n \exp\left(-\frac{\gamma_n |\nabla E_{F,n}| (E_t - E_0)}{kTF}\right) \quad (372)$$

with E_0 and E_t according to Eq. 368 and Eq. 369. p is replaced by an analogous expression.

The parameters $\gamma_n = n/(n + n_{\text{ref}})$ and $\gamma_p = p/(p + p_{\text{ref}})$ are close to one for significantly large densities and vanish for small densities, switching off the density correction and improving numeric robustness. The reference densities n_{ref} and p_{ref} are specified (in cm^{-3}) by the DenCorRef parameter pair in the TrapAssistedTunneling parameter set. They default to 10^3 cm^{-3} .

The parameters for the Schenk model are accessible in the parameter set TrapAssistedTunneling. The default parameters implemented in Sentaurus Device are related to the gold acceptor level: $E_{\text{trap}} = 0 \text{ eV}$, $S = 3.5$, and $\hbar\omega_0 = 0.068 \text{ eV}$. TrapAssistedTunneling provides a parameter MinField (specified in Vcm^{-1}) used for smoothing at small electric fields. A value of zero (the default) disables smoothing.

Hurkx TAT Model

The following equations apply to electrons and holes. The lifetimes and capture cross sections become functions of the trap-assisted tunneling factor $g(F) = \Gamma_{\text{tat}}$:

$$\tau = \tau_0 / (1 + \Gamma_{\text{tat}}) \quad , \quad \sigma = \sigma_0 (1 + \Gamma_{\text{tat}}) \quad (373)$$

where Γ_{tat} is given by:

$$\Gamma_{\text{tat}} = \int_0^{\tilde{E}_n} \exp\left[u - \frac{2}{3} \frac{\sqrt{u^3}}{\tilde{E}}\right] du \quad (374)$$

with the approximate solutions:

$$\Gamma_{\text{tat}} \approx \begin{cases} \sqrt{\pi} \tilde{E} \cdot \exp\left[\frac{1}{3} \tilde{E}^2\right] \left(2 - \text{erfc}\left[\frac{1}{2} \left(\frac{\tilde{E}_n}{\tilde{E}} - \tilde{E}\right)\right]\right), & \tilde{E} \leq \sqrt{\tilde{E}_n} \\ \sqrt{\pi} \tilde{E} \cdot \tilde{E}_n^{1/4} \exp\left[-\tilde{E}_n + \tilde{E} \sqrt{\tilde{E}_n} + \frac{1}{3} \frac{\sqrt{\tilde{E}_n^3}}{\tilde{E}}\right] \text{erfc}[\tilde{E}_n^{1/4} \sqrt{\tilde{E}} - \tilde{E}_n^{3/4} / \sqrt{\tilde{E}}], & \tilde{E} > \sqrt{\tilde{E}_n} \end{cases} \quad (375)$$

where $\tilde{E}$ and $\tilde{E}_n$ are respectively defined as:

$$\tilde{E} = \frac{E}{E_0}, \text{ where } E_0 = \frac{\sqrt{8m_0m_tk^3T^3}}{q\hbar} \quad (376)$$

$$\tilde{E}_n = \frac{E_n}{kT} = \begin{cases} 0, & kT \ln \frac{n}{n_i} > 0.5E_g \\ \frac{0.5E_g}{kT} - \ln \frac{n}{n_i}, & E_{\text{trap}} \leq kT \ln \frac{n}{n_i} \leq 0.5E_g \\ \frac{0.5E_g}{kT} - \frac{E_{\text{trap}}}{kT}, & E_{\text{trap}} > kT \ln \frac{n}{n_i} \end{cases} \quad (377)$$

where m_t is the carrier tunneling mass and E_{trap} is an energy of trap level that is taken from SRH recombination if the model is applied to the lifetimes (τ) or from trap equations if it is applied to cross sections (σ).

When quantization is active (see [Chapter 14 on page 315](#)), the classical density:

$$n_{\text{cl}} = n \exp\left(\frac{\Lambda}{kT}\right) \quad (378)$$

rather than the true density n enters [Eq. 377](#).

The Hurkx model has only one parameter, m_t , the carrier tunneling mass, which can be specified in the parameter file for electrons and holes as follows:

```
HurkxTrapAssistedTunneling {
    mt = <value>, <value>
}
```

Dynamic Nonlocal Path Trap-assisted Tunneling

The local Schenk and Hurkx TAT models described in [SRH Field Enhancement on page 419](#) may fail to give accurate results at low temperatures or near abrupt junctions with a strong, nonuniform, electric field profile. Moreover, the local TAT models are not suitable for computing TAT in heterojunctions where the band edge is modulated by the material composition rather than the electric field. In addition to the local Schenk and Hurkx TAT models, Sentaurus Device provides dynamic nonlocal path Schenk and Hurkx TAT models. These models take into account nonlocal TAT processes with the WKB transmission coefficient based on the exact tunneling barrier. In the present models, electrons and holes are captured in or emitted from the defect level at different locations by the phonon-assisted tunneling process. As a result, the position-dependent electron and hole recombination rates as

well as the recombination rate at the defect level are all different. For each location and for each carrier type, the tunneling path is determined dynamically based on the energy band profile rather than predefined by the nonlocal mesh. Therefore, the present models do not require user-specification of the nonlocal mesh.

NOTE The nonlocal path TAT models are not suitable for AC or noise analysis as nonlocal derivative terms in the Jacobian matrix are not taken into account. The lack of the derivative terms can degrade convergence when the high-field saturation mobility model is switched on, or when the series resistance is defined at electrodes, or when there is a floating region.

Recombination Rate

To compute the net electron recombination rate, the model dynamically searches for the tunneling path with the following assumptions:

- The tunneling path is a straight line with its direction equal to the gradient of the conduction band at the starting position.
- The tunneling energy is equal to the conduction band energy at the starting position and is equal to the defect level ($E_T(x) = E_{\text{trap}}(x) + E_i(x)$) at the ending position.
- Electrons can be captured in or emitted from the defect level at any location between the starting position and the ending position of the tunneling path.
- When the tunneling path encounters Neumann boundaries or semiconductor–insulator interfaces, it undergoes specular reflection.

For a given tunneling path of length l that starts at $x = 0$ and ends at $x = l$, electron capture and emission between the conduction band at $x = 0$ and the defect level at any position x for $0 \leq x \leq l$ is possible. As a result, the net electron recombination rate at $x = 0$ from the conduction band due to the TAT process $R_{\text{net},n}^{\text{TAT}}$ involves integration along the path as follows:

$$R_{\text{net},n}^{\text{TAT}} = C_n \int_0^l \frac{\Gamma_C(x, E_C(0))}{\tau_n(x)} \frac{T(0) + T(x)}{2\sqrt{T(0)T(x)}} \left[\exp\left[\frac{E_{F,n}(0) - E_C(0)}{kT(0)}\right] f^p(x) - \exp\left[\frac{E_T(x) - E_C(0)}{kT(x)}\right] f^n(x) \right] dx \quad (379)$$

$$C_n = |\nabla E_C(0)| \frac{N_C(0)}{kT(0)} \left(\exp\left[\frac{E_{F,n}(0) - E_C(0)}{kT(0)}\right] + 1 \right)^{-\alpha} \quad (380)$$

$$\Gamma_C(x, \epsilon) = W_C(x, \epsilon) \exp\left(-2 \int_0^x \kappa_C(x, \epsilon) dr\right) \quad (381)$$

where:

- τ_n is the electron lifetime.
- $\alpha = 0$ for Boltzmann statistics and $\alpha = 1$ for Fermi–Dirac statistics.
- f^n is the electron occupation probability at the defect level.
- f^p is the hole occupation probability at the defect level.
- κ_C is the magnitude of the imaginary wavevector obtained from the effective mass approximation or from the Franz two-band dispersion relation (Eq. 704, p. 717).
- $W_C(x, \varepsilon) = 1$ for the nonlocal Hurkx TAT model. For the nonlocal Schenk TAT model, $W_C(x, \varepsilon)$ is modeled as:

$$W_C(x, \varepsilon) = \frac{\hbar[E_C(x) - E_C(0)] \exp(0.5) W[\varepsilon - E_T(x)]}{4x[E_C(x) - E_T(x)] \sqrt{\pi m_C k T(x)} W[E_C(x) + kT(x)/2 - E_T(x)]} \quad (382)$$

$$W(\varepsilon) = \frac{1}{\sqrt{\chi}} \exp\left(\frac{\varepsilon}{2kT} + \chi\right) \left(\frac{z}{l + \chi}\right)^l \quad (383)$$

where:

- $\chi = \sqrt{l^2 + z^2}$.
- $l = \varepsilon / \hbar \omega_0$.
- $z = S / \sinh(\hbar \omega_0 / 2kT)$.
- $\hbar \omega_0$ is the phonon energy.
- S is the Huang–Rhys constant.

The net hole recombination rate can be obtained in a similar way. The quasistatic electron and hole occupation probabilities at the defect level can be determined by balancing the net electron capture rate and the net hole capture rate.

Using Dynamic Nonlocal Path TAT Model

The nonlocal path TAT model is switched on when the model is selected in the SRH option as follows:

```
Recombination( ...
  SRH( ...
    NonlocalPath (
      Lifetime = Schenk | Hurkx      # Hurkx model is used by default
      Fermi | -Fermi                # use alpha=1 (alpha=0 by default)
      TwoBand | -TwoBand            # use Franz two band dispersion (off by default)
    )
  )
)
```

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Shockley–Read–Hall Recombination

`Lifetime=Hurkx` (default) selects the nonlocal Hurkx model, while `Lifetime=Schenk` selects the nonlocal Schenk model. The `Fermi` option sets α equal to 1 ($\alpha = 0$ by default). The `TwoBand` option activates the Franz two-band dispersion (Eq. 704, p. 717) instead of the effective mass approximation for the computation of the imaginary wavevector.

The nonlocal path TAT model introduces no additional model parameters. The same minority carrier lifetime for the SRH model is used in the nonlocal path TAT model. The doping- and temperature-dependent lifetime as well as the lifetime loaded from file can be specified together with the nonlocal path TAT model.

NOTE The SRH field-enhancement model should not be used together with the nonlocal path TAT model to avoid double-counting of the TAT process.

The nonlocal Hurkx model uses the tunneling mass specified in the `HurkxTrapAssistedTunneling` parameter set. The tunneling mass, the phonon energy, and the Huang–Rhys coupling constant for the nonlocal Schenk model are specified in the `TrapAssistedTunneling` parameter set.

The maximum length of the tunneling path is determined by the parameter `MaxTunnelLength` in the `Band2BandTunneling` parameter set.

You can consider electron TAT from Schottky contacts or Schottky metal–semiconductor interfaces using the nonlocal path TAT model. To consider the electron capture and emission from the Schottky contact, you must specify the parameter `TATNonlocalPathNC`, which represents the room temperature effective density-of-states ($N_C(300\text{K})$) for the Schottky contact in the electrode-specific `Physics` section, for example:

```
Physics (electrode = "source") { ...  
    TATNonlocalPathNC = 2.0e19  
}
```

Similarly, to consider the TAT from the Schottky metal–semiconductor interface, you must specify the room temperature effective density-of-states in the corresponding interface-specific `Physics` section, for example:

```
Physics (MaterialInterface = "Gold/Silicon") { ...  
    TATNonlocalPathNC = 2.0e19  
}
```

Sentaurus Device assumes that the effective density-of-states at T follows:

$$N_C(T) = N_C(300\text{K}) \left(\frac{T}{300\text{K}} \right)^{\frac{3}{2}} \quad (384)$$

The electron and hole recombination rates as well as the recombination rate at the defect level can be plotted as:

```
Plot { ...
    eSRHRecombination    # electron recombination rate
    hSRHRecombination    # hole recombination rate
    tSRHRecombination    # recombination rate at the defect level
}
```

Trap-assisted Auger Recombination

Trap-assisted Auger (TAA) recombination is a modification to the SRH recombination (see [Shockley–Read–Hall Recombination on page 415](#)) and coupled defect level (see [Coupled Defect Level \(CDL\) Recombination on page 429](#)) models. When TAA is active, Sentaurus Device uses the lifetimes [4]:

$$\frac{\tau_p}{1 + \tau_p/\tau_p^{\text{TAA}}} \quad (385)$$

$$\frac{\tau_n}{1 + \tau_n/\tau_n^{\text{TAT}}} \quad (386)$$

in place of the lifetimes τ_p and τ_n in [Eq. 355](#) and [Eq. 393](#).

The TAA lifetimes in [Eq. 385](#) depend on the carrier densities:

$$\frac{1}{\tau_n^{\text{TAA}}} \approx C_p^{\text{TAA}}(n + p) \quad (387)$$

$$\frac{1}{\tau_p^{\text{TAA}}} \approx C_n^{\text{TAA}}(n + p) \quad (388)$$

A reasonable order of magnitude for the TAA coefficients C_n^{TAA} and C_p^{TAA} is $1 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$ to $1 \times 10^{-11} \text{ cm}^3 \text{ s}^{-1}$; the default values are $C_n^{\text{TAA}} = C_p^{\text{TAA}} = 1 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$.

TAA recombination is activated by using the keyword `TrapAssistedAuger` in the Recombination statement in the Physics section (see [Table 208 on page 1388](#)):

```
Physics {
    Recombination(TrapAssistedAuger ...)
    ...
}
```

The trap-assisted Auger parameters C_n^{TAA} and C_p^{TAA} are accessible in the parameter set `TrapAssistedAuger`.

Surface SRH Recombination

The surface SRH recombination model can be activated at semiconductor–semiconductor and semiconductor–insulator interfaces (see [Interface-specific Models on page 64](#)).

At interfaces, an additional formula is used that is structurally equivalent to the bulk expression of the SRH generation–recombination:

$$R_{\text{surf, net}}^{\text{SRH}} = \frac{np - n_{\text{i,eff}}^2}{(n + n_1)/s_p + (p + p_1)/s_n} \quad (389)$$

with:

$$n_1 = n_{\text{i,eff}} \exp\left(\frac{E_{\text{trap}}}{kT}\right) \text{ and } p_1 = n_{\text{i,eff}} \exp\left(\frac{-E_{\text{trap}}}{kT}\right) \quad (390)$$

For Fermi statistics and quantization, the equations are modified in the same manner as for bulk SRH recombination (see [Eq. 359](#)).

The recombination velocities of otherwise identically prepared surfaces depend, in general, on the concentration of dopants at the surface [8][9][10]. Particularly, in cases where the doping concentration varies along an interface, it is desirable to include such a doping dependence. Sentaurus Device models doping dependence of surface recombination velocities according to:

$$s = s_0 \left[1 + s_{\text{ref}} \left(\frac{N_i}{N_{\text{ref}}} \right)^\gamma \right] \quad (391)$$

The results of Cuevas [10] indicate that for phosphorus-diffused silicon, $\gamma = 1$; while the results of King and Swanson [9] seem to imply that no significant doping dependence exists for the recombination velocities of boron-diffused silicon surfaces.

To activate the model, specify the option `SurfaceSRH` to the `Recombination` keyword in the `Physics` section for the respective interface. To plot the surface recombination, specify `SurfaceRecombination` in the `Plot` section. The parameters E_{trap} , s_{ref} , N_{ref} , and γ are accessible in the parameter set `SurfaceRecombination`. The corresponding values for silicon are given in [Table 72 on page 429](#).

Table 72 Surface SRH parameters

Symbol	Parameter name	Electrons	Holes	Unit
s_0	S0	1×10^3	1×10^3	cm/s
s_{ref}	Sref	1×10^{-3}		1
N_{ref}	Nref	1×10^{16}		cm^{-3}
γ	gamma	1		1
E_{trap}	Etrap	0		eV

The doping dependence of the recombination velocity can be suppressed by setting s_{ref} to zero.

NOTE The `SurfaceSRH` keyword also is supported for metal–semiconductor interfaces, but the model it activates there is different from the one described in this section (see [Electric Boundary Conditions for Metals on page 274](#)).

Coupled Defect Level (CDL) Recombination

The steady state recombination rate for two coupled defect levels generalizes the familiar single-level SRH formula. An important feature of the model is a possibly increased field effect that may lead to large excess currents. The model is discussed in the literature [11].

Using CDL

The CDL recombination can be switched on using the keyword `CDL` in the `Physics` section:

```
Physics{ Recombination( CDL ... ) ... }
```

The contributions R_1 and R_2 in Eq. 394 can be plotted by using the keywords `CDL1` and `CDL2` in the `Plot` section. For the net rate and coupling term, $R - R_1 - R_2$, the keywords `CDL` and `CDL3` must be specified.

CDL Model

The notation of the model parameters is illustrated in [Figure 31](#).

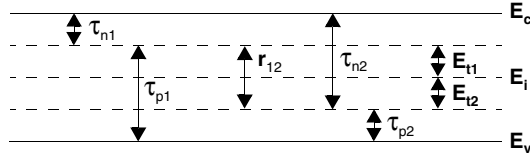


Figure 31 Notation for CDL recombination including all capture and emission processes

The CDL recombination rate is given by:

$$R = R_1 + R_2 + \left(\sqrt{R_{12}^2 - S_{12}} - R_{12} \right) \times \frac{\tau_{n1}\tau_{p2}(n + n_2)(p + p_1) - \tau_{n2}\tau_{p1}(n + n_1)(p + p_2)}{r_1 r_2} \quad (392)$$

with:

$$r_j = \tau_{nj}(p + p_j) + \tau_{pj}(n + n_j) \quad (393)$$

$$R_j = \frac{np - n_{i,eff}^2}{r_j}, \quad j = 1, 2 \quad (394)$$

$$R_{12} = \frac{r_1 r_2}{2r_{12}\tau_{n1}\tau_{n2}\tau_{p1}\tau_{p2}(1 - \varepsilon)} + \frac{\tau_{n1}(p + p_1) + \tau_{p2}(n + n_2)}{2\tau_{n1}\tau_{p2}(1 - \varepsilon)} + \frac{\varepsilon[\tau_{n2}(p + p_2) + \tau_{p1}(n + n_1)]}{2\tau_{n2}\tau_{p1}(1 - \varepsilon)} \quad (395)$$

$$S_{12} = \frac{1}{\tau_{n1}\tau_{p2}(1 - \varepsilon)} \left(1 - \frac{\tau_{n1}\tau_{p2}}{\tau_{n2}\tau_{p1}} \varepsilon \right) (np - n_{i,eff}^2) \quad (396)$$

$$\varepsilon = \exp\left(-\frac{|E_{t2} - E_{t1}|}{kT}\right) \quad (397)$$

where τ_{ni} and τ_{pi} denote the electron and hole lifetimes of the defect level i . The coupling parameter between the defect levels is called r_{12} (keyword `TrapTrapRate` in the parameter file). The carrier lifetimes are calculated analogously to the carrier lifetimes in the SRH model (see [Shockley–Read–Hall Recombination on page 415](#)). The number of parameters is doubled compared to the SRH model, and they are changeable in the parameter set CDL.

The quantities n_2 and p_2 are the corresponding quantities of n_1 and p_1 for the second defect level. They are defined analogously to [Eq. 356, p. 415](#) and [Eq. 357, p. 415](#).

For Fermi statistics and quantization, the equations are modified in the same manner as for SRH recombination (see [Eq. 359](#), p. 416).

Radiative Recombination

Using Radiative Recombination

The radiative recombination model is activated in the `Physics` section by the keyword `Radiative`:

```
Physics {
    Recombination (Radiative)
}
```

It can also be switched on or off by using the notation `+Radiative` or `-Radiative`:

```
Physics (Region = "gate") {
    Recombination (+Radiative)
}

Physics (Material = "AlGaAs") {
    Recombination (-Radiative)
}
```

The value of the radiative recombination rate is plotted as follows:

```
Plot {
    RadiativeRecombination
}
```

The value of the parameter C can be changed in the parameter file:

```
RadiativeRecombination {
    C = 2.5e-10
}
```

Radiative Model

The radiative (direct) recombination model expresses the recombination rate as:

$$R_{\text{net}} = C \cdot (np - n_{i,\text{eff}}^2) \quad (398)$$

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Auger Recombination

By default, Sentaurus Device selects $C = 2 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ for GaAs and $C = 0 \text{ cm}^3 \text{ s}^{-1}$ for other materials. For Fermi statistics and quantization, the equations are modified in the same manner than for SRH recombination (see [Eq. 359, p. 416](#)).

Auger Recombination

The rate of band-to-band Auger recombination R_{net}^A is given by:

$$R_{\text{net}}^A = (C_n n + C_p p)(np - n_{i,\text{eff}}^2) \quad (399)$$

with temperature-dependent Auger coefficients [\[12\]\[13\]\[14\]](#):

$$C_n(T) = \left(A_{A,n} + B_{A,n} \left(\frac{T}{T_0} \right) + C_{A,n} \left(\frac{T}{T_0} \right)^2 \right) \left[1 + H_n \exp \left(-\frac{n}{N_{0,n}} \right) \right] \quad (400)$$

$$C_p(T) = \left(A_{A,p} + B_{A,p} \left(\frac{T}{T_0} \right) + C_{A,p} \left(\frac{T}{T_0} \right)^2 \right) \left[1 + H_p \exp \left(-\frac{p}{N_{0,p}} \right) \right] \quad (401)$$

where $T_0 = 300 \text{ K}$. There is experimental evidence for a decrease of the Auger coefficients at high injection levels [\[14\]](#). This effect is explained as resulting from exciton decay: at lower carrier densities, excitons, which are loosely bound electron–hole pairs, increase the probability for Auger recombination. Excitons decay at high carrier densities, resulting in a decrease of recombination. This effect is modeled by the terms $1 + H \exp(-n/N_0)$ in [Eq. 400](#) and [Eq. 401](#).

Auger recombination is typically important at high carrier densities. Therefore, this injection dependence will only be seen in devices where extrinsic recombination effects are extremely low, such as high-efficiency silicon solar cells. The injection dependence of the Auger coefficient can be deactivated by setting H to zero in the parameter file.

For Fermi statistics and quantization, the equations are modified in the same manner as for SRH recombination (see [Eq. 359, p. 416](#)). Default values for silicon are listed in [Table 73](#).

Table 73 Default coefficients of Auger recombination model

Symbol	$A_A [\text{cm}^6 \text{ s}^{-1}]$	$B_A [\text{cm}^6 \text{ s}^{-1}]$	$C_A [\text{cm}^6 \text{ s}^{-1}]$	$H [1]$	$N_0 [\text{cm}^{-3}]$
Parameter name	A	B	C	H	N0
Electrons	6.7×10^{-32}	2.45×10^{-31}	-2.2×10^{-32}	3.46667	1×10^{18}
Holes	7.2×10^{-32}	4.5×10^{-33}	2.63×10^{-32}	8.25688	1×10^{18}

Auger recombination is activated with the argument `Auger` in the `Recombination` statement:

```
Physics{ Recombination( Auger ...) ...}
```

By default, Sentaurus Device uses [Eq. 399](#) only if R_{net}^A is positive and replaces the value by zero if R_{net}^A is negative. To use [Eq. 399](#) for negative values (that is, to allow for Auger generation of electron–hole pairs), use the `WithGeneration` option to the `Auger` keyword:

```
Physics { Recombination( Auger(WithGeneration) ...) ...}
```

The Auger parameters are accessible in the parameter set `Auger`.

Constant Carrier Generation

The simplest generation model computes a constant carrier generation G_{const} . It is activated in the `Physics` section as follows:

```
Physics {
  Recombination (
    ConstantCarrierGeneration (value = 1e10)    # [cm^-3 s^-1]
  )
}
```

Alternatively, the value of G_{const} can be specified in the parameter file:

```
ConstantCarrierGeneration {
  value = 1e10          # [cm^-3 s^-1]
}
```

You can ramp the generation rate by specifying a corresponding `Goal` in a `Quasistationary` command:

```
Goal { Model = "ConstantCarrierGeneration"
  Parameter = "value"
  Value = 1e20}
```

To visualize G_{const} , plot the value of `PMIRecombination`:

```
Plot {
  PMIRecombination
}
```

The model also can be specified regionwise or materialwise.

NOTE The constant carrier generation model is functionally equivalent to the constant optical generation model presented in [Constant Optical Generation on page 548](#).

Avalanche Generation

Electron–hole pair production due to avalanche generation (impact ionization) requires a certain threshold field strength and the possibility of acceleration, that is, wide space charge regions. If the width of a space charge region is greater than the mean free path between two ionizing impacts, charge multiplication occurs, which can cause electrical breakdown. The reciprocal of the mean free path is called the ionization coefficient α . With these coefficients for electrons and holes, the generation rate can be expressed as:

$$G_{ii} = \alpha_n n v_n + \alpha_p p v_p \quad (402)$$

Sentaurus Device implements several models for the ionization coefficients: van Overstraeten–de Man, Okuto–Crowell, Lackner, University of Bologna, the new University of Bologna, and Hatakeyama.

Sentaurus Device allows you to select the appropriate driving force for the simulation, that is, the method used to compute the accelerating field. The choices include GradQuasiFermi, Eparallel, CarrierTempDrive, and ElectricField (see [Driving Force on page 445](#)).

Using Avalanche Generation

Avalanche generation is switched on by using the keyword `Avalanche` in the `Recombination` statement in the `Physics` section. The keywords `eAvalanche` and `hAvalanche` specify separate models or driving forces for the electron and hole ionization coefficients, respectively.

The model is selected by using one of the keywords `vanOverstraeten`, `Okuto`, `Lackner`, `UniBo`, `UniBo2`, or `Hatakeyama`. The default model is `vanOverstraeten`.

The driving force is selected using one of the keywords `GradQuasiFermi`, `Eparallel`, `CarrierTempDrive`, or `ElectricField`. The default driving force is `GradQuasiFermi`.

For example:

```
Physics {  
    Recombination(eAvalanche(CarrierTempDrive) hAvalanche(Okuto) ... )  
}
```

selects the default van Overstraeten–de Man model for the electron impact ionization process with a driving force derived from electron temperature, and selects the Okuto–Crowell model for holes using the default driving force based on GradQuasiFermi.

To include a dependency on energy bandgap in the avalanche generation models, specify the keyword `BandgapDependence` as an argument to `Avalanche`, `eAvalanche`, or `hAvalanche` (see the model descriptions in the next sections). For example:

```
Physics {
    Recombination(Avalanche(Lackner BandgapDependence))
}
```

To plot the avalanche generation rate, specify `AvalancheGeneration` in the `Plot` section. To plot either of the two terms on the right-hand side of Eq. 402 separately, specify `eAvalanche` or `hAvalanche`. To plot α_n or α_p , specify `eAlphaAvalanche` or `hAlphaAvalanche`, respectively.

van Overstraeten – de Man Model

This model is based on the Chynoweth law [15]:

$$\alpha(F_{\text{ava}}) = \gamma a \exp\left(-\frac{\gamma b}{F_{\text{ava}}}\right) \quad (403)$$

with:

$$\gamma = \frac{\tanh\left(\frac{\hbar\omega_{\text{op}}}{2kT_0}\right)}{\tanh\left(\frac{\hbar\omega_{\text{op}}}{2kT}\right)} \quad (404)$$

The factor γ with the optical phonon energy $\hbar\omega_{\text{op}}$ expresses the temperature dependence of the phonon gas against which carriers are accelerated. The coefficients a , b , and $\hbar\omega_{\text{op}}$, as measured by van Overstraeten and de Man [16], are applicable over the range of fields $1.75 \times 10^5 \text{ Vcm}^{-1}$ to $6 \times 10^5 \text{ Vcm}^{-1}$ and are listed in Table 74 on page 436.

Two sets of coefficients a and b are used for high and low ranges of electric field. The values $a(\text{low})$, $b(\text{low})$ are used in the low field range up to E_0 and the values $a(\text{high})$, $b(\text{high})$ apply in the high field above E_0 . For electrons, the impact ionization coefficients are by default the same in both field ranges.

If `BandgapDependence` is specified as an argument to `Avalanche`, the coefficient b in Eq. 403 is replaced with:

$$b \rightarrow \frac{\beta E_g}{q\lambda} \quad (405)$$

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Avalanche Generation

where E_g is the energy bandgap, λ is the optical-phonon mean free path for the carrier, and β is a proportionality constant. Default values for λ and β are listed in [Table 74](#).

You can adjust the coefficient values in the Sentaurus Device parameter set `vanOverstraetendeMan`.

Table 74 Coefficients for van Overstraeten–de Man model ([Eq. 403](#)) for silicon

Symbol	Parameter name	Electrons	Holes	Valid range of electric field	Unit
a	a (low)	7.03×10^5	1.582×10^6	$1.75 \times 10^5 \text{ Vcm}^{-1}$ to E_0	cm^{-1}
	a (high)	7.03×10^5	6.71×10^5	E_0 to $6 \times 10^5 \text{ Vcm}^{-1}$	
b	b (low)	1.231×10^6	2.036×10^6	$1.75 \times 10^5 \text{ Vcm}^{-1}$ to E_0	V/cm
	b (high)	1.231×10^6	1.693×10^6	E_0 to $6 \times 10^5 \text{ Vcm}^{-1}$	
E_0	E0	4×10^5	4×10^5		V/cm
$\hbar\omega_{\text{op}}$	hbarOmega	0.063	0.063		eV
λ	lambda	62×10^{-8}	45×10^{-8}		cm
β	beta (low)	0.678925	0.815009	$1.75 \times 10^5 \text{ Vcm}^{-1}$ to E_0	1
	beta (high)	0.678925	0.677706	E_0 to $6 \times 10^5 \text{ Vcm}^{-1}$	

Okuto–Crowell Model

Okuto and Crowell [\[17\]](#) suggested the empirical model:

$$\alpha(F_{\text{ava}}) = a \cdot (1 + c(T - T_0)) F_{\text{ava}}^\gamma \exp \left[- \left(\frac{b[1 + d(T - T_0)]}{F_{\text{ava}}} \right)^\delta \right] \quad (406)$$

where $T_0 = 300 \text{ K}$ and the user-adjustable coefficients are listed in [Table 75](#) with their default values for silicon. These values apply to the range of electric field from 10^5 Vcm^{-1} to 10^6 Vcm^{-1} .

If `BandgapDependence` is specified as an argument to `Avalanche`, the coefficient b in [Eq. 406](#) is replaced with:

$$b \rightarrow \frac{\beta E_g}{q\lambda} \quad (407)$$

where E_g is the energy bandgap, λ is the optical-phonon mean free path for the carrier, and β is a proportionality constant. Default values for λ and β are listed in [Table 75 on page 437](#).

You can adjust the coefficient values in the parameter set Okuto.

Table 75 Coefficients for Okuto–Crowell model (Eq. 406) for silicon

Symbol	Parameter name	Electrons	Holes	Unit
a	a	0.426	0.243	V^{-1}
b	b	4.81×10^5	6.53×10^5	V/cm
c	c	3.05×10^{-4}	5.35×10^{-4}	K^{-1}
d	d	6.86×10^{-4}	5.67×10^{-4}	K^{-1}
γ	gamma	1	1	1
δ	delta	2	2	1
λ	lambda	62×10^{-8}	45×10^{-8}	cm
β	beta	0.265283	0.261395	1

Lackner Model

Lackner [18] derived a pseudo-local ionization rate in the form of a modification to the Chynoweth law, assuming stationary conditions. The temperature-dependent factor γ was introduced to the original model:

$$\alpha_v(F_{\text{ava}}) = \frac{\gamma a_v}{Z} \exp\left(-\frac{\gamma b_v}{F_{\text{ava}}}\right) \text{ where } v = n, p \quad (408)$$

with:

$$Z = 1 + \frac{\gamma b_n}{F_{\text{ava}}} \exp\left(-\frac{\gamma b_n}{F_{\text{ava}}}\right) + \frac{\gamma b_p}{F_{\text{ava}}} \exp\left(-\frac{\gamma b_p}{F_{\text{ava}}}\right) \quad (409)$$

and:

$$\gamma = \frac{\tanh\left(\frac{\hbar\omega_{\text{op}}}{2kT_0}\right)}{\tanh\left(\frac{\hbar\omega_{\text{op}}}{2kT}\right)} \quad (410)$$

The default values of the coefficients a , b , and $\hbar\omega_{\text{op}}$ are applicable in silicon for the range of the electric field from 10^5 Vcm^{-1} to 10^6 Vcm^{-1} .

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Avalanche Generation

If `BandgapDependence` is specified as an argument to `Avalanche`, the b coefficients in [Eq. 408](#) and [Eq. 409](#) are replaced with:

$$b \rightarrow \frac{\beta E_g}{q\lambda} \quad (411)$$

where E_g is the energy bandgap, λ is the optical-phonon mean free path for the carrier, and β is a proportionality constant.

[Table 76](#) lists the default model parameters. The model parameters are accessible in the parameter set `Lackner`.

Table 76 Coefficients for Lackner model ([Eq. 408](#)) for silicon

Symbol	Parameter name	Electrons	Holes	Unit
a	a	1.316×10^6	1.818×10^6	cm^{-1}
b	b	1.474×10^6	2.036×10^6	V/cm
$\hbar\omega_{\text{op}}$	hbarOmega	0.063	0.063	eV
λ	lambda	62×10^{-8}	45×10^{-8}	cm
β	beta	0.812945	0.815009	1

University of Bologna Impact Ionization Model

The University of Bologna impact ionization model was developed for an extended temperature range between 25°C and 400°C (see also [New University of Bologna Impact Ionization Model on page 440](#) for an updated version of this model). It is based on impact ionization data generated by the Boltzmann solver HARM [\[19\]](#). It covers a wide range of electric fields (50 kVcm^{-1} to 600 kVcm^{-1}) and temperatures (300 K to 700 K). It is calibrated against impact ionization measurements [\[20\]\[21\]](#) in the whole temperature range.

The model reads:

$$\alpha(F_{\text{ava}}, T) = \frac{F_{\text{ava}}}{a(T) + b(T) \exp\left[\frac{d(T)}{F_{\text{ava}} + c(T)}\right]} \quad (412)$$

The temperature dependence of the model parameters, determined by fitting experimental data, reads (for electrons):

$$a(T) = a_0 + a_1 t^{a_2} \quad b(T) = b_0 \quad c(T) = c_0 + c_1 t + c_2 t^2 \quad d(T) = d_0 + d_1 t + d_2 t^2 \quad (413)$$

and for holes:

$$a(T) = a_0 + a_1 t \quad b(T) = b_0 \exp[b_1 t] \quad c(T) = c_0 t^{c_1} \quad d(T) = d_0 + d_1 t + d_2 t^2 \quad (414)$$

where $t = T/1 \text{ K}$.

If BandgapDependence is specified as an argument to Avalanche, then $d(T)$ is modified as follows:

$$d(T) \rightarrow \left(\frac{\beta E_g}{q \lambda} / d_0 \right) (d_0 + d_1 t + d_2 t^2) \quad (415)$$

where E_g is the energy bandgap, λ is the optical-phonon mean free path for the carrier, and β is a proportionality constant.

Table 77 lists the default model parameters. The model parameters are accessible in the parameter set UniBo.

Table 77 Coefficients for University of Bologna impact ionization model for silicon

Symbol	Parameter name	Electrons	Holes	Unit
a_0	ha0	4.3383	2.376	V
a_1	ha1	-2.42×10^{-12}	1.033×10^{-2}	V
a_2	ha2	4.1233	0	1
b_0	hb0	0.235	0.17714	V
b_1	hb1	0	-2.178×10^{-3}	1
c_0	hc0	1.6831×10^4	9.47×10^{-3}	Vcm^{-1}
c_1	hc1	4.3796	2.4924	Vcm^{-1} , 1
c_2	hc2	0.13005	0	Vcm^{-1} , 1
d_0	hd0	1.2337×10^6	1.4043×10^6	Vcm^{-1}
d_1	hd1	1.2039×10^3	2.9744×10^3	Vcm^{-1}
d_2	hd2	0.56703	1.4829	Vcm^{-1}
λ	lambda	62×10^{-8}	45×10^{-8}	cm
β	beta	0.680414	0.562140	1

NOTE When the University of Bologna impact ionization model is selected for both carriers, that is, `Recombination(Avalanche(UniBo))`, Sentaurus Device also enables Auger generation (see [Auger Recombination on page 432](#)). Specify `Auger(-WithGeneration)` to disable this generation term if necessary.

New University of Bologna Impact Ionization Model

The impact ionization model described in [University of Bologna Impact Ionization Model on page 438](#) was developed further in [22][23][24] to cover an extended temperature range between 25°C and 500°C (773 K). It is based on impact ionization data generated by the Boltzmann solver HARM [19] and is calibrated against specially designed impact ionization measurements [20][21]. It covers a wide range of electric fields. The model reads:

$$\alpha(F_{\text{ava}}, T) = \frac{F_{\text{ava}}}{a(T) + b(T) \exp\left[\frac{d(T)}{F_{\text{ava}} + c(T)}\right]} \quad (416)$$

where the coefficients a , b , c , and d are polynomials of T :

$$a(T) = \sum_{k=0}^3 a_k \left(\frac{T}{1\text{K}}\right)^k \quad (417)$$

$$b(T) = \sum_{k=0}^{10} b_k \left(\frac{T}{1\text{K}}\right)^k \quad (418)$$

$$c(T) = \sum_{k=0}^3 c_k \left(\frac{T}{1\text{K}}\right)^k \quad (419)$$

$$d(T) = \sum_{k=0}^3 d_k \left(\frac{T}{1\text{K}}\right)^k \quad (420)$$

If `BandgapDependence` is specified as an argument to `Avalanche`, $d(T)$ is modified as follows:

$$d(T) \rightarrow \left(\frac{\beta E_g}{q\lambda} / d_0\right) \sum_{k=0}^3 d_k \left(\frac{T}{1\text{K}}\right)^k \quad (421)$$

where E_g is the energy bandgap, λ is the optical-phonon mean free path for the carrier, and β is a proportionality constant.

Table 78 lists the default model parameters. The model parameters are accessible in the parameter set UniBo2.

Table 78 Coefficients for UniBo2 impact ionization model for silicon

Symbol	Electrons		Holes		Unit
	Parameter Name	Value	Parameter Name	Value	
a_0	a0_e	4.65403	a0_h	2.26018	V
a_1	a1_e	-8.76031×10^{-3}	a1_h	0.0134001	V
a_2	a2_e	1.34037×10^{-5}	a2_h	-5.87724×10^{-6}	V
a_3	a3_e	-2.75108×10^{-9}	a3_h	-1.14021×10^{-9}	V
b_0	b0_e	-0.128302	b0_h	0.058547	V
b_1	b1_e	4.45552×10^{-3}	b1_h	-1.95755×10^{-4}	V
b_2	b2_e	-1.0866×10^{-5}	b2_h	2.44357×10^{-7}	V
b_3	b3_e	9.23119×10^{-9}	b3_h	-1.33202×10^{-10}	V
b_4	b4_e	-1.82482×10^{-12}	b4_h	2.68082×10^{-14}	V
b_5	b5_e	-4.82689×10^{-15}	b5_h	0	V
b_6	b6_e	1.09402×10^{-17}	b6_h	0	V
b_7	b7_e	-1.24961×10^{-20}	b7_h	0	V
b_8	b8_e	7.55584×10^{-24}	b8_h	0	V
b_9	b9_e	-2.28615×10^{-27}	b9_h	0	V
b_{10}	b10_e	2.73344×10^{-31}	b10_h	0	V
c_0	c0_e	7.76221×10^3	c0_h	1.95399×10^4	Vcm ⁻¹
c_1	c1_e	25.18888	c1_h	-104.441	Vcm ⁻¹
c_2	c2_e	-1.37417×10^{-3}	c2_h	0.498768	Vcm ⁻¹
c_3	c3_e	1.59525×10^{-4}	c3_h	0	Vcm ⁻¹
d_0	d0_e	7.10481×10^5	d0_h	2.07712×10^6	Vcm ⁻¹
d_1	d1_e	3.98594×10^3	d1_h	993.153	Vcm ⁻¹
d_2	d2_e	-7.19956	d2_h	7.77769	Vcm ⁻¹

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Avalanche Generation

Table 78 Coefficients for UniBo2 impact ionization model for silicon

Symbol	Electrons		Holes		Unit
	Parameter Name	Value	Parameter Name	Value	
d_3	d3_e	6.96431×10^{-3}	d3_h	0	Vcm^{-1}
λ	lambda_e	62×10^{-8}	lambda_h	45×10^{-8}	cm
β	beta_e	0.391847	beta_h	0.831470	1

NOTE The name of the UniBo2 model is case sensitive. Both in the command file and the parameter file, the exact spelling of UniBo2 must be used.

Hatakeyama Avalanche Model

The Hatakeyama avalanche model has been proposed [29] to describe the anisotropic behavior in 4H-SiC power devices. The impact ionization coefficient α is obtained according to the Chynoweth law [15]:

$$\alpha = \gamma a e^{\frac{\gamma b}{F}} \quad (422)$$

with:

$$\gamma = \frac{\tanh\left(\frac{\hbar\omega_{\text{op}}}{2kT_0}\right)}{\tanh\left(\frac{\hbar\omega_{\text{op}}}{2kT}\right)} \quad (423)$$

The coefficients a and b are computed dependent on the direction of the driving force $\vec{F}$. For this purpose, the driving force $\vec{F}$ is decomposed into a component F_{0001} parallel to the anisotropic axis 0001 and a remaining component $F_{11\bar{2}0}$ perpendicular to the anisotropic axis.

The norm $F = \|\vec{F}\|_2$ satisfies the equation:

$$F^2 = F_{0001}^2 + F_{11\bar{2}0}^2 \quad (424)$$

Based on the two projections F_{0001} and $F_{11\bar{2}0}$, the coefficients a and b then are computed as follows:

$$B = \frac{F}{\sqrt{\left(\frac{F_{11\bar{2}0}}{b_{11\bar{2}0}}\right)^2 + \left(\frac{F_{0001}}{b_{0001}}\right)^2}} \quad (425)$$

$$a = a_{11\bar{2}0} \left(\frac{BF_{11\bar{2}0}}{b_{11\bar{2}0}F} \right)^2 a_{0001} \left(\frac{BF_{0001}}{b_{0001}F} \right)^2 \quad (426)$$

$$A = \log \frac{a_{0001}}{a_{11\bar{2}0}} \quad (427)$$

$$b = B \sqrt{1 - \theta A^2 \left(\frac{BF_{11\bar{2}0}F_{0001}}{Fb_{11\bar{2}0}b_{0001}} \right)^2} \quad (428)$$

With the default $\theta = 1$, the coefficient b may become undefined for large values of the driving force F (the argument of the square root in Eq. 428 becomes negative). This can only happen if:

$$F > \frac{2\min(b_{0001}, b_{11\bar{2}0})}{\left| \log \frac{a_{0001}}{a_{11\bar{2}0}} \right|} \quad (429)$$

By setting $\theta = 0$, you have $b = B$, and the coefficients a and b only depend on the direction of the driving force $\vec{F}$, but not its magnitude.

Table 79 shows the default parameters for 4H-SiC.

Table 79 4H-SiC coefficients for Hatakeyama model

Symbol	Parameter name	Electrons	Holes	Unit
a_{0001}	a_0001	1.76×10^8	3.41×10^8	cm^{-1}
$a_{11\bar{2}0}$	a_1120	2.10×10^7	2.96×10^7	cm^{-1}
b_{0001}	b_0001	3.30×10^7	2.50×10^7	V/cm
$b_{11\bar{2}0}$	b_1120	1.70×10^7	1.60×10^7	V/cm
$\hbar\omega_{\text{op}}$	hbarOmega	0.19	0.19	eV
θ	theta	1	1	1

Driving Force

In contrast to other avalanche models in Sentaurus Device (see [Driving Force on page 445](#)), which only use the magnitude $F_{\vec{F}}$ of the driving force, the Hatakeyama avalanche model requires a vectorial driving force $\vec{F}$ to compute the coefficients a and b . Depending on the driving force model, the following expressions are used:

- **ElectricField:** The driving force $\vec{F}$ is defined as the straight electric field $\vec{E}$.

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Avalanche Generation

- `Eparallel`: The driving force $\vec{F}$ is given by (where $\vec{j}$ is the electron or hole current density):

$$\vec{F} = \frac{\vec{j} \cdot \vec{T}}{\|\vec{j}\|} E \quad (430)$$

- `GradQuasiFermi`: The driving force $\vec{F}$ is given by (where Φ is the electron or hole quasi-Fermi potential):

$$\vec{F} = \nabla \Phi \quad (431)$$

- `CarrierTempDrive`: The driving force $\vec{F}$ is given by (where E^{eff} is the effective field obtained from the electron or hole carrier temperature):

$$\vec{F} = \frac{\vec{j} \cdot \vec{T}}{\|\vec{j}\|} E^{\text{eff}} \quad (432)$$

See also [Avalanche Generation With Hydrodynamic Transport on page 445](#).

Anisotropic Coordinate System

In a 2D simulation, Sentaurus Device assumes that the y-axis in the crystal system is the anisotropic axis 0001, and the x-axis in the crystal system is the isotropic axis 11 $\bar{2}$ 0. In a 3D simulation, the z-axis in the crystal system is the anisotropic axis, and the x-axis and y-axis span the isotropic plane.

The simulation coordinate system relative to the crystal system is defined by the `x` and `y` vectors in the `LatticeParameters` section of the parameter file (see [Crystal and Simulation Coordinate Systems on page 767](#) and [Coordinate Systems on page 870](#)).

Usage

The Hatakeyama avalanche model uses a special-purpose interpolation formula to compute the coefficients a and b based on the direction of the driving force. This formula differs from the standard approach as described in [Anisotropic Avalanche Generation on page 777](#).

NOTE The Hatakeyama avalanche model must not be used together with an `Aniso(Avalanche)` specification in the `Physics` section.

Driving Force

In Sentaurus Device, the driving force F_{ava} for impact ionization can be computed as the component of the electrostatic field in the direction of the current, $F_{\text{ava}} = \vec{F} \cdot \hat{J}_{n,p}$, (keyword `Eparallel`), or the value of the gradient of the quasi-Fermi level, $F_{\text{ava}} = |\nabla \Phi_{n,p}|$, (keyword `GradQuasiFermi`). For these two possibilities, F_{ava} is affected by the keyword `ParallelToInterfaceInBoundaryLayer` (see [Field Correction Close to Interfaces on page 401](#)). For hydrodynamic simulations, F_{ava} can be computed from the carrier temperature (keyword `CarrierTempDrive`, see [Avalanche Generation With Hydrodynamic Transport on page 445](#)). The default model is `GradQuasiFermi` and, only for hydrodynamic simulations, it is `CarrierTempDrive`. See [Table 209 on page 1388](#) for a summary of keywords.

The option `ElectricField` is used to perform breakdown simulations using the ‘ionization integral’ method (see [Approximate Breakdown Analysis on page 446](#)).

Avalanche Generation With Hydrodynamic Transport

If the hydrodynamic transport model is used, the default driving force F_{ava} equals an effective field E^{eff} obtained from the carrier temperature. The usual conversion of local carrier temperatures to effective fields E^{eff} is described by the algebraic equations:

$$n\mu_n(E_n^{\text{eff}})^2 = n\frac{3kT_n - T}{2q\lambda_n\tau_{en}} \quad (433)$$

$$p\mu_p(E_p^{\text{eff}})^2 = p\frac{3kT_p - T}{2q\lambda_p\tau_{ep}} \quad (434)$$

which are obtained from the energy conservation equation under time-independent, homogeneous conditions. [Eq. 433](#) and [Eq. 434](#) have been simplified in Sentaurus Device by using the assumption $\mu_n E_n^{\text{eff}} = v_{\text{sat},n}$ and $\mu_p E_p^{\text{eff}} = v_{\text{sat},p}$. This assumption is true for high values of the electric field. However, for low field, the impact ionization rate is negligibly small.

The parameters λ_n and λ_p are fitting coefficients (default value 1) and their values can be changed in the parameter set `AvalancheFactors`, where they are represented as `n_l_f` and `p_l_f`, respectively.

The conventional conversion formulas [Eq. 433](#) and [Eq. 434](#) can be activated by specifying parameters $\Upsilon_n = 0$, $\Upsilon_p = 0$ in the same `AvalancheFactors` section.

The simplified conversion formulas discussed above predict a linear dependence of effective electric field on temperature for high values of carrier temperature. For silicon, however, Monte Carlo simulations do not confirm this behavior. To obtain a better agreement with Monte Carlo data, additional heat sinks must be taken into account by the inclusion of an additional term in the equations for E^{eff} .

Such heat sinks arise from nonelastic processes, such as the impact ionization itself. Sentaurus Device supports the following model to account for these heat sinks:

$$n\nu_{\text{sat},n}E_n^{\text{eff}} = n\frac{3kT_n - T}{2q\lambda_n\tau_{en}} + \frac{\Upsilon_n}{q}(E_g + \delta_n kT_n)\alpha_n n\nu_{\text{sat},n} \quad (435)$$

A similar equation E_p^{eff} is used to determine E_p^{eff} . To activate this model, set the parameters Υ_n and Υ_p to 1. This is the default for silicon, where the generalized conversion formula [Eq. 435](#) gives good agreement with Monte Carlo data for $\delta_n = \delta_p = 3/2$. For all other materials, the default of the parameters Υ_n and Υ_p is 0.

Table 80 Hydrodynamic avalanche model: Default parameters

Symbol	Parameter name	Default value
λ_n	n_l_f	1
λ_p	p_l_f	1
Υ_n	n_gamma	1
Υ_p	p_gamma	1
δ_n	n_delta	1.5
δ_p	p_delta	1.5

NOTE This procedure ensures that the same results are obtained as with the conventional local field–dependent models in the bulk case. Conversely, the temperature-dependent impact ionization model usually gives much more accurate predictions for the substrate current in short-channel MOS transistors.

Approximate Breakdown Analysis

Junction breakdown due to avalanche generation is simulated by inspecting the ionization integrals:

$$I_n = \int_0^W \alpha_n(x) e^{-\int_x^W (\alpha_n(x') - \alpha_p(x')) dx'} dx \quad (436)$$

$$I_p = \int_0^W \alpha_p(x) e^{-\int_0^x (\alpha_p(x') - \alpha_n(x')) dx'} dx \quad (437)$$

where α_n , α_p are the ionization coefficients for electrons and holes, respectively, and W is the width of the depletion zone. The integrations are performed along field lines through the depletion zone. Avalanche breakdown occurs if an ionization integral equals one. [Eq. 436](#) describes electron injection (electron primary current) and [Eq. 437](#) describes hole injection. Since these breakdown criteria do not depend on current densities, a breakdown analysis can be performed by computing only the Poisson equation and ionization integrals under the assumption of constant quasi-Fermi levels in the depletion region.

Using Breakdown Analysis

To enable breakdown analysis, Sentaurus Device provides the driving force `ElectricField`, which can be computed even for constant quasi-Fermi levels. This driving force is less physical than the others available in Sentaurus Device. Therefore, Synopsys discourages using it for any purpose other than approximate breakdown analysis.

`ComputeIonizationIntegrals` in the `Math` section switches on the computation of the ionization integrals for ionization paths crossing the local field maxima in the semiconductor. By default, Sentaurus Device reports only the path with the largest I_{mean} . With the addition of the keyword `WriteAll`, information about the ionization integrals for all computed paths is written to the log file.

The `Math` keyword `BreakAtIonIntegral` is used to terminate the quasistationary simulation when the largest ionization integral is greater than one.

The complete syntax of this keyword is `BreakAtIonIntegral(<number> <value>)` where a quasistationary simulation finishes if the number ionization integral is greater than value, and the ionization integrals are ordered with respect to decreasing value:

```
Math { BreakAtIonIntegral }
```

Three optional keywords in the `Plot` section specify the values of the corresponding ionization integrals that are stored along the breakdown paths:

```
Plot { eIonIntegral | hIonIntegral | MeanIonIntegral }
```

These ionization integrals can be visualized using Sentaurus Visual.

A typical command file of Sentaurus Device is:

```
Electrode {
  { name="anode" Voltage=0 }
  { name="cathode" Voltage=600 }
}
File {
  grid      = "@grid@"
  doping    = "@doping@"
  current   = "@plot@"
  output    = "@log@"
  plot      = "@data@"
}
Physics {
  Mobility (DopingDependence HighFieldSaturation)
  Recombination(SRH Auger Avalanche(ElectricField))
}
Solve {
  Quasistationary(
    InitialStep=0.02 MaxStep=0.01 MinStep=0.01
    Goal {name=cathode voltage=1000}
  )
  { poisson }
}
Math {
  Iterations=100
  BreakAtIonIntegral
  ComputeIonizationIntegrals(WriteAll)
}
Plot {
  eIonIntegral hIonIntegral MeanIonIntegral
  eDensity hDensity
  ElectricField/Vector
  eAlphaAvalanche hAlphaAvalanche
}
```

Approximate Breakdown Analysis With Carriers

Sentaurus Device allows ionization integrals to be calculated when solving the electron and hole current continuity equations. In this case, however, impact ionization-generated carriers will be included self-consistently in the solution, and the benefits of performing an approximate breakdown analysis (faster simulations with fewer convergence issues) will be lost.

To prevent the self-consistent inclusion of impact ionization–generated carriers in the solution of the device equations, specify the option `AvalPostProcessing` in the `Math` section:

```
Math { AvalPostProcessing }
```

This option allows ionization integrals to be calculated even when solving for carriers, but retains the benefits of an approximate breakdown analysis.

Band-to-Band Tunneling Models

Sentaurus Device provides several band-to-band tunneling models:

- Schenk model (see [Schenk Model on page 451](#)).
- Hurkx model (see [Hurkx Band-to-Band Model on page 453](#)).
- A family of simple models (see [Simple Band-to-Band Models on page 452](#)).
- Nonlocal path model (see [Dynamic Nonlocal Path Band-to-Band Model on page 454](#)).

The Schenk, Hurkx, and simple models use a common approach to suppress artificial band-to-band tunneling near insulator interfaces and for rapidly varying fields (see [Tunneling Near Interfaces and Equilibrium Regions on page 454](#)).

Using Band-to-Band Tunneling

Band-to-band tunneling is controlled by the `Band2Band` option of `Recombination`:

```
Recombination( ...
  Band2Band (
    Model = Schenk | Hurkx | E1 | E1_5 | E2 | NonlocalPath
    DensityCorrection = Local | None
    InterfaceReflection | -InterfaceReflection
    FranzDispersion | -FranzDispersion
    ParameterSetName = (<string>...)
  )
)
```

`Model` determines the model:

- `Schenk` selects the Schenk model.
- `Hurkx` selects the Hurkx model.
- `E1`, `E1_5`, and `E2` select one of the simple models.
- `NonlocalPath` selects the nonlocal path model.

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Band-to-Band Tunneling Models

`DensityCorrection` is used by the Schenk, Hurkx, and simple models, and its default value is `None`. A value of `Local` activates a local-density correction (see [Schenk Density Correction on page 452](#)).

`InterfaceReflection` is used by the nonlocal path model, and this option is switched on by default. This option allows a tunneling path reflected at semiconductor–insulator interfaces. When it is switched off, band-to-band tunneling is neglected when the tunneling path encounters semiconductor–insulator interfaces.

`FranzDispersion` is used by the nonlocal path model, and this option is switched off by default. When it is switched on, the Franz dispersion relation (Eq. 704, p. 717) instead of the Kane dispersion relation (Eq. 446, p. 456) for the imaginary wavevector is used in the direct tunneling process. The indirect tunneling process is not affected by this option.

All models use `ParameterSetName`. It specifies a list of names of `Band2BandTunneling` parameter sets. For each name, band-to-band tunneling is computed using the parameters in the named parameter set, and the results are all added to give the total band-to-band tunneling rate. Without `ParameterSetName`, only the band-to-band tunneling obtained with the unnamed `Band2BandTunneling` parameter set is computed. For example:

```
Band2Band(  
    Model=NonlocalPath  
    ParameterSetName=("phonon-assisted" "direct")  
)
```

will use the nonlocal path model, summing the contributions obtained with the `phonon-assisted` and `direct` parameter sets.

For backward compatibility:

- `Band2Band` alone (without parameters) selects the Schenk model with local-density correction.
- `Band2Band(Hurkx)` selects the Hurkx model without density correction.
- `Band2Band(E1)`, `Band2Band(E1_5)`, and `Band2Band(E2)` select one of the simple models.

The parameters for all band-to-band models are available in the `Band2BandTunneling` parameter set. The parameters specific to individual models are described in the respective sections. The parameter `MinField` (specified in Vcm^{-1}) is used by the Schenk, Hurkx, and simple models for smoothing at small electric fields. A value of zero (the default) disables smoothing.

Named `Band2BandTunneling` parameter sets are specified in the parameter file, for example:

```
Band2BandTunneling "phonon-assisted" { ... }
```

which specifies a parameter set with the name `phonon-assisted`. Note that parameters in named `Band2BandTunneling` parameter sets do not have default values, so you must specify values for all the parameters used by the model you select in the command file.

Schenk Model

Phonon-assisted band-to-band tunneling cannot be neglected in steep p-n junctions (with a doping level of $1 \times 10^{19} \text{ cm}^{-3}$ or more on both sides) or in high normal electric fields of MOS structures. It must be switched on if the field, in some regions of the device, exceeds (approximately) $8 \times 10^5 \text{ V/cm}$. In this case, defect-assisted tunneling (see [SRH Field Enhancement on page 419](#)) must also be switched on.

Band-to-band tunneling is modeled using the expression [25]:

$$R_{\text{net}}^{\text{bb}} = AF^{7/2} \frac{\tilde{n}\tilde{p} - n_{i,\text{eff}}^2}{(\tilde{n} + n_{i,\text{eff}})(\tilde{p} + n_{i,\text{eff}})} \left[\frac{(F_C^{\mp})^{-3/2} \exp\left(-\frac{F_C^{\mp}}{F}\right)}{\exp\left(\frac{\hbar\omega}{kT}\right) - 1} + \frac{(F_C^{\pm})^{-3/2} \exp\left(-\frac{F_C^{\pm}}{F}\right)}{1 - \exp\left(-\frac{\hbar\omega}{kT}\right)} \right] \quad (438)$$

where $\tilde{n}$ and $\tilde{p}$ equal n and p for `DensityCorrection=None`, and are given by [Eq. 440](#) for `DensityCorrection=Local`. The critical field strengths read:

$$F_C^{\pm} = B(E_{g,\text{eff}} \pm \hbar\omega)^{3/2} \quad (439)$$

The upper sign in [Eq. 438](#) refers to tunneling generation ($np < n_{i,\text{eff}}^2$) and the lower sign refers to recombination ($np > n_{i,\text{eff}}^2$). The quantity $\hbar\omega$ denotes the energy of the transverse acoustic phonon.

For Fermi statistics and quantization, [Eq. 438](#) is modified in the same way as for SRH recombination (see [Eq. 359, p. 416](#)).

The parameters [25] are given in [Table 81](#) and can be accessed in the parameter set `Band2BandTunneling`. The defaults were obtained assuming the field direction to be $\langle 111 \rangle$.

Table 81 Coefficients for band-to-band tunneling (Schenk model)

Symbol	Parameter name	Default value	Unit
A	A	8.977×10^{20}	$\text{cm}^{-1} \text{s}^{-1} \text{V}^{-2}$
B	B	2.14667×10^7	$\text{Vcm}^{-1} \text{eV}^{-3/2}$
$\hbar\omega$	hbarOmega	18.6	meV

Schenk Density Correction

The modified electron density reads:

$$\tilde{n} = n \left(\frac{n_{i,eff}}{N_C} \right)^{\frac{\gamma_n |\nabla E_{F,n}|}{F}} \quad (440)$$

and there is a similar relation for $\tilde{p}$. The parameters $\gamma_n = n/(n + n_{ref})$ and $\gamma_p = p/(p + p_{ref})$ work as discussed in [Schenk TAT Density Correction on page 422](#). The reference densities n_{ref} and p_{ref} are specified (in cm^{-3}) by the `DenCorRef` parameter pair in the `Band2BandTunneling` parameter set. An additional parameter `MinGradQF` (specified in eV cm^{-1}) is available for smoothing at small values of the gradient for the Fermi potential.

Simple Band-to-Band Models

Sentaurus Device provides a family of simple band-to-band models. Compared to advanced models, the most striking weakness of the simple models is that they predict a nonzero generation rate even in equilibrium. A general expression for these models can be written for the generation term [26] as:

$$G^{b2b} = AF^P \exp\left(-\frac{B}{F}\right) \quad (441)$$

Depending on the value of `Model`, P takes the value 1, 1.5, or 2.

[Table 82](#) lists the coefficients of models and their defaults. The coefficients A and B can be changed in the parameter set `Band2BandTunneling`.

Table 82 Coefficients for band-to-band tunneling (simple models)

Model	P	A	B
E1	1	$1.1 \times 10^{27} \text{ cm}^{-2} \text{ s}^{-1} \text{ V}^{-1}$	$21.3 \times 10^6 \text{ V cm}^{-1}$
E1_5	1.5	$1.9 \times 10^{24} \text{ cm}^{-1.5} \text{ s}^{-1} \text{ V}^{-1.5}$	$21.9 \times 10^6 \text{ V cm}^{-1}$
E2	2	$3.4 \times 10^{21} \text{ cm}^{-1} \text{ s}^{-1} \text{ V}^{-2}$	$22.6 \times 10^6 \text{ V cm}^{-1}$

Hurkx Band-to-Band Model

Similar to the other band-to-band tunneling models, in the Hurkx model [27], the tunneling carriers are modeled by an additional generation–recombination process. Its contribution is expressed as:

$$R_{\text{net}}^{\text{bb}} = A \cdot D \cdot \left(\frac{F}{1 \text{ V/cm}} \right)^P \exp \left(- \frac{BE_g(T)^{3/2}}{E_g(300\text{K})^{3/2} F} \right) \quad (442)$$

where:

$$D = \frac{np - n_{\text{i,eff}}^2}{(n + n_{\text{i,eff}})(p + n_{\text{i,eff}})} (1 - |\alpha|) + \alpha \quad (443)$$

Here, specifying $\alpha = 0$ gives the original Hurkx model, whereas $\alpha = -1$ gives only generation ($D = -1$), and $\alpha = 1$ gives only recombination ($D = 1$). Therefore, if $D < 0$, it is a net carrier generation model. If $D > 0$, it is a recombination model. For Fermi statistics and quantization, Eq. 443 is modified in the same way as for SRH recombination (see Eq. 359).

For `DensityCorrection=Local`, n and p in Eq. 443 are replaced by $\tilde{n}$ and $\tilde{p}$ (see [Schenk Density Correction on page 452](#)).

The coefficients A (in $\text{cm}^{-3}\text{s}^{-1}$), B (in V/cm), P , and α can be specified in the `Band2BandTunneling` parameter set. By default, Sentaurus Device uses $\alpha = 0$ and the parameters from the E2 model (see [Table 82 on page 452](#)). Different values for the generation (A_{gen} , B_{gen} , P_{gen}) and recombination (A_{rec} , B_{rec} , P_{rec}) of carriers are supported. For example, to change the parameters to those used in [27], use:

```
Band2BandTunneling {
    Agen = 4e14 # [1/(cm³s)]
    Bgen = 1.9e7 # [V/cm]
    Pgen = 2.5 # [1]
    Arec = 4e14 # [1/(cm³s)]
    Brec = 1.9e7 # [V/cm]
    Prec = 2.5 # [1]
    alpha = 0 # [1]
}
```

Tunneling Near Interfaces and Equilibrium Regions

Physically, band-to-band tunneling occurs over a certain tunneling distance. If the material properties or the electric field change significantly over this distance, [Eq. 438](#), [Eq. 441](#), and [Eq. 442](#) become inaccurate. In particular, near insulator interfaces, band-to-band tunneling vanishes, as no states to tunnel to are available in the insulator.

In some parts of the device (near equilibrium regions), it is possible that the electric field is large but changes rapidly, so that the actual tunneling distance (the distance over which the electrostatic potential change amounts to the band gap) is bigger and, therefore, tunneling is much smaller than expected from the local field alone.

To account for these effects, two additional control parameters are introduced in the `Band2BandTunneling` parameter set:

```
dDist = <value> # [cm]
dPot  = <value> # [V]
```

By default, both these parameters equal zero. Sentaurus Device disables band-to-band tunneling within a distance `dDist` from insulator interfaces. If `dPot` is nonzero (reasonable values for `dPot` are of the order of the band gap), Sentaurus Device disables band-to-band tunneling at each point where the change of the electrostatic potential in field direction within a distance of $dPot/F$ is smaller than `dPot/2`.

Dynamic Nonlocal Path Band-to-Band Model

Sufficient band-bending caused by electric fields or heterostructures can make electrons in the valence band valley (Γ -valley in the k -space), at a certain location, reach the conduction band valley (Γ -, X -, or L -valley in the k -space) at different locations using direct or phonon-assisted band-to-band tunneling process.

The present model implements the nonlocal generation of electrons and holes caused by direct and phonon-assisted band-to-band tunneling processes [\[28\]](#). In direct semiconductors such as GaAs and InAs, the direct tunneling process is usually dominant. On the other hand, the phonon-assisted tunneling process is dominant in indirect semiconductors such as Si and Ge. If the energy differences between the conduction band valleys are small, it is possible that both the direct and the phonon-assisted tunneling processes are important.

The generation rate is obtained from the nonlocal path integration, and electrons and holes are generated nonlocally at the ends of the tunneling path. As a result, the position-dependent electron and hole generation rates are different in the present model. The model can be applied to heterostructure devices with abrupt and graded heterojunctions.

The main difference between the present model and the band-to-band tunneling model based on the nonlocal mesh (see [Band-to-Band Contributions to Nonlocal Tunneling Current on page 721](#)) is that the tunneling path is determined dynamically based on the energy band profile rather than predefined by the nonlocal mesh. Therefore, the present model does not require user-specification of the nonlocal mesh.

The model dynamically searches for the tunneling path with the following assumptions:

- The tunneling path starts from the valence band in a region where the nonlocal path model is active.
- The tunneling path is a straight line with its direction opposite to the gradient of the valence band at the starting position.
- The tunneling energy is equal to the valence band energy at the starting position and is equal to the conduction band energy plus band offset at the ending position.
- When the tunneling path encounters Neumann boundaries or semiconductor–insulator interfaces, it undergoes specular reflection.
- The tunneling path ends at the conduction band.

If the path crosses a region where the nonlocal path model is not active, by default, the tunneling from this path is discarded. If the `-eB2BGenWithinSelectedRegions` flag is specified in the global `Math` section, tunneling for all paths entirely within semiconductor regions are accounted for.

NOTE By default, nonlocal derivative terms in the Jacobian matrix are not taken into account. To use AC or noise analysis with the present model, computation of nonlocal derivatives must be switched on (see [Using Nonlocal Path Band-to-Band Model on page 458](#)). The lack of derivative terms can degrade convergence when the high-field saturation mobility model is switched on or the series resistance is defined at electrodes.

Band-to-Band Generation Rate

For a given tunneling path of length l that starts at $x = 0$ and ends at $x = l$, holes are generated at $x = 0$ and electrons are generated at $x = l$. The net hole recombination rate at $x = 0$ due to the direct band-to-band tunneling process R_{net}^d can be written as:

$$R_{\text{net}}^d = |\nabla E_V(0)| C_d \exp\left(-2 \int_0^l \kappa dx\right) \left[\left(\exp\left[\frac{\mathcal{E} - E_{F,n}(l)}{kT(l)}\right] + 1 \right)^{-1} - \left(\exp\left[\frac{\mathcal{E} - E_{F,p}(0)}{kT(0)}\right] + 1 \right)^{-1} \right] \quad (444)$$

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Band-to-Band Tunneling Models

$$C_d = \frac{g\pi}{36h} \left(\int_0^l \frac{dx}{\kappa} \right)^{-1} \left[1 - \exp \left(-k_m^2 \int_0^l \frac{dx}{\kappa} \right) \right] \quad (445)$$

where:

- h is Planck's constant.
- g is the degeneracy factor.
- $\epsilon = E_V(0) = E_C(l) + \Delta_C(l)$ is the tunneling energy.
- Δ_C is the conduction band offset. Δ_C can be positive if the considered tunneling process involves the conduction band valley whose energy minimum is greater than the conduction band energy E_C .
- κ is the magnitude of the imaginary wavevector obtained from the Kane two-band dispersion relation [28]:

$$\kappa = \frac{1}{\hbar} \sqrt{m_r E_{g,\text{tun}} (1 - \alpha^2)} \quad (446)$$

$$\alpha = -\frac{m_0}{2m_r} + 2 \sqrt{\frac{m_0}{2m_r} \left(\frac{\epsilon - E_V}{E_{g,\text{tun}}} - \frac{1}{2} \right) + \frac{m_0^2}{16m_r^2} + \frac{1}{4}} \quad (447)$$

$$\frac{1}{m_r} = \frac{1}{m_V} + \frac{1}{m_C} \quad (448)$$

$E_{g,\text{tun}} = E_{g,\text{eff}} + \Delta_C$ is the effective band gap including the band offset, and k_m is the maximum transverse momentum determined by the maximum valence-band energy ϵ_{max} and the minimum conduction-band energy ϵ_{min} :

$$k_m^2 = \min(k_{\text{vm}}^2, k_{\text{cm}}^2) \quad (449)$$

$$k_{\text{vm}}^2 = \frac{2m_V(\epsilon_{\text{max}} - \epsilon)}{\hbar^2} \quad (450)$$

$$k_{\text{cm}}^2 = \frac{2m_C(\epsilon - \epsilon_{\text{min}})}{\hbar^2} \quad (451)$$

In the Kane two-band dispersion relation, the effective mass in the conduction band m_C and the valence band m_V are related [28]:

$$\frac{1}{m_C} = \frac{1}{2m_r} + \frac{1}{m_0} \quad (452)$$

$$\frac{1}{m_V} = \frac{1}{2m_r} - \frac{1}{m_0} \quad (453)$$

The net hole recombination rate due to the phonon-assisted band-to-band tunneling process R_{net}^p can be written as:

$$R_{\text{net}}^p = |\nabla E_V(0)| C_p \exp \left(-2 \int_0^{x_0} \kappa_V dx - 2 \int_{x_0}^l \kappa_C dx \right) \left[\left(\exp \left[\frac{\varepsilon - E_{F,n}(l)}{kT(l)} \right] + 1 \right)^{-1} - \left(\exp \left[\frac{\varepsilon - E_{F,p}(0)}{kT(0)} \right] + 1 \right)^{-1} \right] \quad (454)$$

$$C_p = \int_0^l \frac{g(1 + 2N_{\text{op}}) D_{\text{op}}^2}{2^6 \pi^2 \rho \varepsilon_{\text{op}} E_{g,\text{tun}}} \sqrt{\frac{m_V m_C}{\hbar l \sqrt{2m_r E_{g,\text{tun}}}}} dx \left(\int_0^{x_0} \frac{dx}{\kappa_V} \right)^{-1} \left(\int_{x_0}^l \frac{dx}{\kappa_C} \right)^{-1} \left[1 - \exp \left(-k_{\text{vm}}^2 \int_0^{x_0} \frac{dx}{\kappa_V} \right) \right] \left[1 - \exp \left(-k_{\text{cm}}^2 \int_{x_0}^l \frac{dx}{\kappa_C} \right) \right] \quad (455)$$

where D_{op} , ε_{op} , and $N_{\text{op}} = [\exp(\varepsilon_{\text{op}}/kT) - 1]^{-1}$ are the deformation potential, energy, and number of optical phonons, respectively, ρ is the mass density, and κ_V and κ_C are the magnitude of the imaginary wavevectors from the Keldysh dispersion relation:

$$\kappa_V = \frac{1}{\hbar} \sqrt{2m_V |\varepsilon - E_V|} \Theta(\varepsilon - E_V) \quad (456)$$

$$\kappa_C = \frac{1}{\hbar} \sqrt{2m_C |E_C + \Delta_C - \varepsilon|} \Theta(E_C + \Delta_C - \varepsilon) \quad (457)$$

and x_0 is the location where $\kappa_V = \kappa_C$.

As Eq. 444, p. 455 and Eq. 454 are the extension of the results in [28] to arbitrary band profiles, these expressions are reduced to the well-known Kane and Keldysh models in the uniform electric-field limit [28]:

$$R_{\text{net}} = A \left(\frac{F}{F_0} \right)^P \exp \left(-\frac{B}{F} \right) \quad (458)$$

where $F_0 = 1 \text{ V/cm}$, $P = 2$ for the direct tunneling process, and $P = 2.5$ for the phonon-assisted tunneling process.

At $T = 300 \text{ K}$ without the bandgap narrowing effect, the prefactor A and the exponential factor B for the direct tunneling process can be expressed by [28]:

$$A = \frac{g \pi m_r^{1/2} (qF_0)^2}{9 \hbar^2 [E_g(300\text{K}) + \Delta_C]^{1/2}} \quad (459)$$

$$B = \frac{\pi^2 m_r^{1/2} [E_g(300\text{K}) + \Delta_C]^{3/2}}{qh} \quad (460)$$

For the phonon-assisted tunneling process, A and B can be expressed by [28]:

$$A = \frac{g(m_V m_C)^{3/2} (1 + 2N_{op}) D_{op}^2 (qF_0)^{5/2}}{2^{21/4} h^{5/2} m_r^{5/4} \rho \epsilon_{op} [E_g(300K) + \Delta_C]^{7/4}} \quad (461)$$

$$B = \frac{2^{7/2} \pi m_r^{1/2} [E_g(300K) + \Delta_C]^{3/2}}{3qh} \quad (462)$$

Using Nonlocal Path Band-to-Band Model

The parameters for the nonlocal path band-to-band model are available in the parameter set `Band2BandTunneling`. Two input parameter sets are selectable: First, when directly specifying m_C , m_V , g , and gD_{op}^2/ρ , Sentaurus Device computes the prefactor A and the exponential factor B . Second, A and B can be chosen as input. In addition, the conduction band offset Δ_C , the phonon energy ϵ_{op} , and the effective mass ratio m_V/m_C can be specified.

Specifying $\epsilon_{op} = 0$ selects the direct tunneling process. Choosing input parameters A , B , and Δ_C determines g and m_r from Eq. 459, p. 457 and Eq. 460, p. 457. Choosing input parameters m_C , m_V , g , and Δ_C determines A and B .

When $\epsilon_{op} = 0$ and the option `FranzDispersion` is switched on in the `Band2Band` option of the command file, the magnitude of the imaginary wavevector is obtained from the Franz two-band dispersion relation (Eq. 704, p. 717) instead of the Kane two-band dispersion relation (Eq. 446, p. 456). In this case, g and m_r are still obtained from Eq. 459 and Eq. 460. Then, Eq. 448, p. 456 and m_V/m_C determine m_V and m_C .

Specifying $\epsilon_{op} > 0$ selects the phonon-assisted tunneling process. Choosing input parameters A , B , Δ_C , ϵ_{op} , and m_V/m_C determines gD_{op}^2/ρ , m_V , and m_C from Eq. 461 and Eq. 462. When $m_V/m_C = 0$, m_V and m_C are determined from Eq. 452, p. 456 and Eq. 453, p. 457. Choosing input parameters m_C , m_V , gD_{op}^2/ρ , Δ_C , and ϵ_{op} determines A and B .

The nonlocal path band-to-band model is activated by setting `Model=NonlocalPath` in the `Band2Band` option of the command file. Multiple processes with different parameters are supported by using the `ParameterSetName` option and multiple named `Band2BandTunneling` parameter sets (see Using Band-to-Band Tunneling on page 449).

NOTE Each tunneling path (characterized by a particular parameter set name) must have a consistent tunneling process (either direct or phonon-assisted tunneling process) in different regions. For example, specifying `Ppath=0` (direct tunneling) in silicon regions and specifying `Ppath` greater than zero (phonon-assisted tunneling) in polysilicon regions will induce an error message.

MaxTunnelLength in the parameter file specifies the maximum length of the tunneling path. If the length reaches MaxTunnelLength before a valid tunneling path is found, the band-to-band tunneling is neglected.

In the parameter file, the parameter pairs QuantumPotentialFactor and QuantumPotentialPosFac in the Band2BandTunneling parameter set specify the prefactors for the electron and hole quantum potentials obtained from the density gradient model (see [Density Gradient Quantization Model on page 326](#)) that can be added to the effective band gap for tunneling. By default, they are zero, such that the quantum potentials are neglected in the computation of the generation rate. When they are nonzero, the quantum potentials multiplied by the corresponding prefactors are added to the conduction and valence band edges when the transmission coefficient is computed. For QuantumPotentialFactor, an addition is performed irrespective of the sign of the quantum potential. For QuantumPotentialPosFac, an addition is performed only when the quantum potential is positive, that is, only where quantization causes an effective widening of the band gap. For example, the following section causes the electron and hole quantum potentials to be added to the band gap wherever they cause an effective widening of the band gap:

```
Band2BandTunneling {
    ...
    QuantumPotentialPosFac = 1 1
}
```

The computation of nonlocal derivative terms can be switched on and off in the global Math section using the Derivative option within the NonlocalPath keyword, for example:

```
Math {
    ...
    NonLocalPath(Derivative=1)
}
```

[Table 83](#) lists the coefficients of models and their defaults. These parameters can be mole fraction dependent. The default parameters *A* and *B* are obtained from [\[27\]](#).

Table 83 Default parameters for nonlocal path band-to-band tunneling model

Symbol	Parameter name	Default value	Unit
<i>A</i>	Apath	4×10^{14}	$\text{cm}^{-3}\text{s}^{-1}$
<i>B</i>	Bpath	1.9×10^7	Vcm^{-1}
gD_{op}^2/ρ	Cpath	0	$\text{J}^2\text{cmkg}^{-1}$
<i>g</i>	degeneracy	0	1
Δ_{C}	Dpath	0	eV
m_{C}	m_c	0	m_0
m_{V}	m_v	0	m_0

16: Generation–Recombination

Bimolecular Recombination

Table 83 Default parameters for nonlocal path band-to-band tunneling model

Symbol	Parameter name	Default value	Unit
ϵ_{op}	Ppath	0.037	eV
m_V/m_C	Rpath	0	1

Visualizing Nonlocal Band-to-Band Generation Rate

To plot the electron and hole generation rates, specify `eBand2BandGeneration` and `hBand2BandGeneration` in the Plot section, respectively.

NOTE `Band2BandGeneration` is equal to `hBand2BandGeneration`.

Bimolecular Recombination

The bimolecular recombination model describes the interaction of electron–hole pairs and singlet excitons (see [Singlet Exciton Equation on page 269](#)).

Physical Model

The rate of electron–hole pair and singlet exciton recombination follows the Langevin form, that is, it is proportional to the carrier mobility. The bimolecular recombination rate is given by:

$$R_{\text{bimolec}} = \gamma \cdot \frac{q}{\epsilon_0 \epsilon_r} \cdot (\mu_n + \mu_p) \left(np - n_{i,\text{eff}}^2 \frac{n_{\text{se}}}{n_{\text{se}}^{\text{eq}}} \right) \quad (463)$$

where:

- γ is a prefactor for the singlet exciton.
- q is the elementary charge.
- ϵ_0 and ϵ_r denote the free space and relative permittivities, respectively.
- Electron and hole mobilities are given by μ_n and μ_p , accordingly.
- n , p , and $n_{i,\text{eff}}$ describe the electron, hole, and effective intrinsic density, respectively.
- n_{se} is the singlet exciton density.
- $n_{\text{se}}^{\text{eq}}$ denotes the singlet-exciton equilibrium density.

Using Bimolecular Recombination

The bimolecular recombination model is activated by using the keyword `Bimolecular` as an argument of the `Recombination` statement in the `SingletExciton` section (see [Table 227 on page 1399](#)). It is switched off by default and can be activated regionwise (see [Singlet Exciton Equation on page 269](#)). An example is:

```
Physics (Region="EML-ETL") {
  SingletExciton (
    Recombination ( Bimolecular )
  )
}
```

[Table 84](#) lists the parameter of the bimolecular recombination model, which is accessible in the `SingletExciton` section of the parameter file.

Table 84 Default parameter for bimolecular recombination

Symbol	Parameter name	Default value	Unit
γ	gamma	0.25	1

Exciton Dissociation Model

The exciton dissociation model describes the dissociation of singlet excitons into electron–hole pairs at semiconductor–semiconductor and semiconductor–insulator interfaces.

Physical Model

The rate of singlet exciton interface dissociation is modeled as:

$$R_{\text{se,diss}}^{\text{surf}} = v_{\text{se},0} \sigma_{\text{se}-N_{\text{diss}}} N_{\text{se,diss}}^{\text{surf}} (n_{\text{se}} - n_{\text{se}}^{\text{eq}}) \quad (464)$$

where:

$$v_{\text{diss}}^{\text{surf}} = v_{\text{se},0} \sigma_{\text{se}-N_{\text{diss}}} N_{\text{se,diss}}^{\text{surf}} \quad (465)$$

is the singlet exciton recombination velocity in cm/s, $\sigma_{\text{se}-N_{\text{diss}}}$ is the capture cross-section of exciton dissociation centers with the surface density $N_{\text{se,diss}}^{\text{surf}}$, $v_{\text{se},0}$ is the singlet exciton thermal velocity, and n_{se} and $n_{\text{se}}^{\text{eq}}$ are the exciton and equilibrium exciton densities, respectively.

In the dissociation process, a singlet exciton generates an electron–hole pair. The rate in [Eq. 464](#) is a recombination rate for the singlet exciton equation and a generation rate for the electron and hole continuity equations.

Using Exciton Dissociation

The exciton dissociation model is activated using the keyword `Dissociation` as an argument of the `Recombination` statement in the `SingletExciton` section (see [Table 227 on page 1399](#)). It is switched off by default and can be activated regionwise (see [Singlet Exciton Equation on page 269](#)).

The following example activates exciton dissociation at the EML/ETL region interface:

```
Physics (RegionInterface="EML/ETL") {
    SingletExciton (Recombination ( Dissociation ))
}
```

[Table 85](#) lists the parameters of the exciton dissociation model, which is accessible in the `SingletExciton` section of the parameter file.

Table 85 Default parameters for exciton dissociation model

Symbol	Parameter name	Default value	Unit
$\sigma_{se-N_{diss}}$	ex_dXsection	1×10^{-8}	cm^2
$N_{se,diss}^{surf}$	N_diss	1×10^{10}	cm^{-2}

References

- [1] D. J. Roulston, N. D. Arora, and S. G. Chamberlain, “Modeling and Measurement of Minority-Carrier Lifetime versus Doping in Diffused Layers of n+-p Silicon Diodes,” *IEEE Transactions on Electron Devices*, vol. ED-29, no. 2, pp. 284–291, 1982.
- [2] J. G. Fossum, “Computer-Aided Numerical Analysis of Silicon Solar Cells,” *Solid-State Electronics*, vol. 19, no. 4, pp. 269–277, 1976.
- [3] J. G. Fossum and D. S. Lee, “A Physical Model for the Dependence of Carrier Lifetime on Doping Density in Nondegenerate Silicon,” *Solid-State Electronics*, vol. 25, no. 8, pp. 741–747, 1982.
- [4] J. G. Fossum *et al.*, “Carrier Recombination and Lifetime in Highly Doped Silicon,” *Solid-State Electronics*, vol. 26, no. 6, pp. 569–576, 1983.